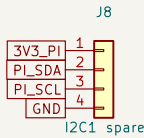
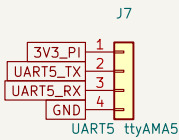
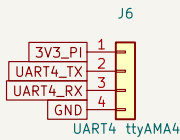
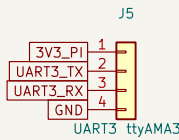
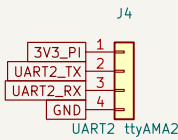
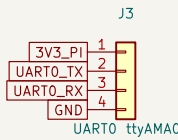
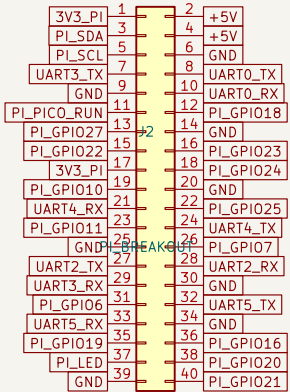
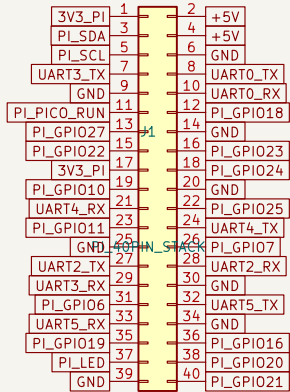




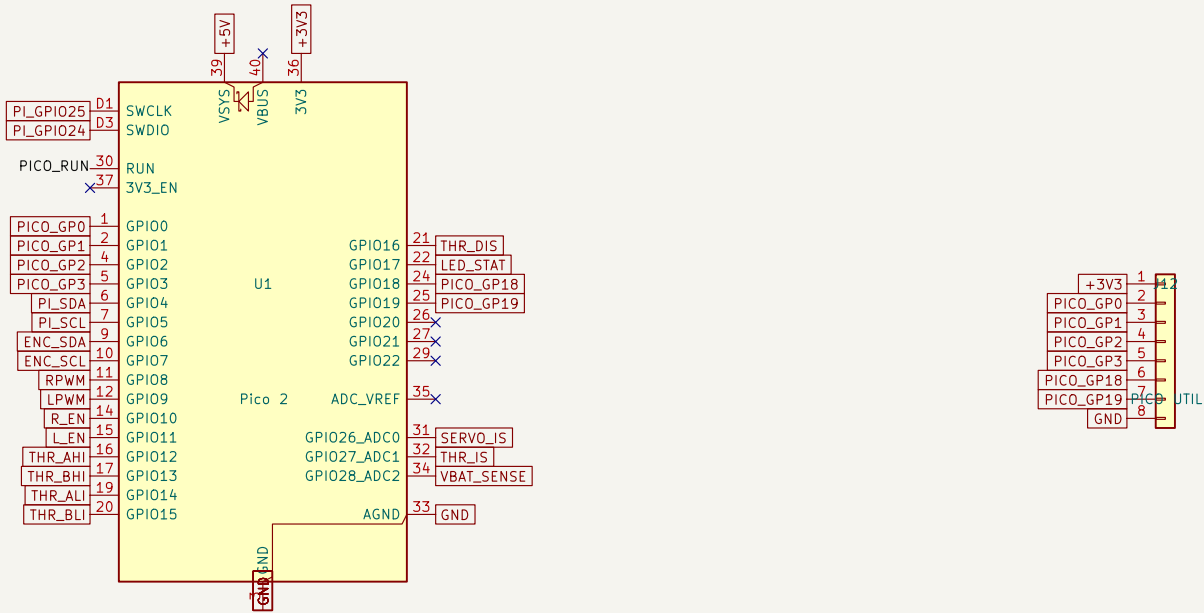


RASPBERRY PI 4/5 CARRIER: stacking header, UART breakouts, display power

DSI touchscreen: ribbon to Pi directly; 5V from J10. USB/HDMI on Pi itself.



MCU: Pico 2 (RP2350) – I2C0 slave @0x42 to Pi, I2C1 master to AS5600
Firmware map: 0x00 CMD {pwm,dir,steer} / 0x10 STATUS {angle,ok,wrap,state,vbat,lsense}; 800ms watchdog
Pico soldered directly (low profile, sits under the Pi). Pi reflashes it via SWD on GPIO24/25 (openocd).



1k



4.7k



4.7k



100k



6.8k



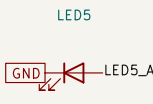
100n



BAT54S

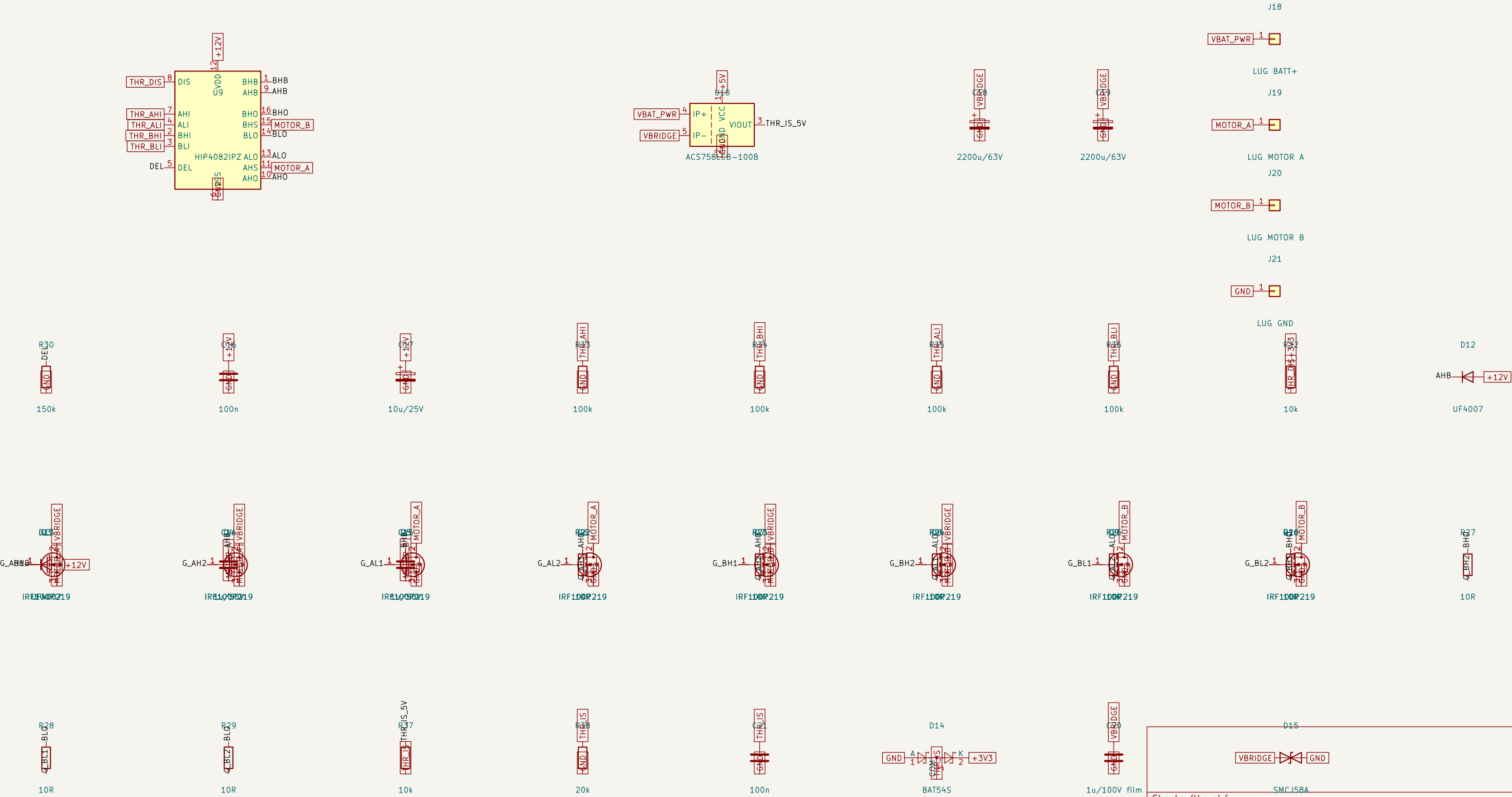


1k



yellow

THRUST: on-board H-bridge, >=800W @ 12V (67A cont), fwd/rev, 12-48V bus
HIP4082 + 8x IRF100P219 (2 per switch) + ACS758-100B battery-leg sense; M5 lugs
Bridge disabled unless Pico drives it: 100k pull-downs on inputs, DIS pulled up to 3V3
Heatsink bar across the FET row required; airflow recommended above 40A continuous



SERVO: on-board 2x BTN8982TA half-bridge (BTS7960-compatible interface), VM = +12V rail
AS5600 (in servo housing per vanchor-ng cad/) cabled to J11; keep lead <1m, 100kHz
24-48V boats: servo stall limited by BUCK2 (4.5A). 12V boats: J17 links VIN->12V, full stall current

